
REPORTING UNDER UNCERTAINTY: HEDGING AND CERTAINTY IN EARLY NEWS COVERAGE

FINAL PROJECT PAPER

Anushay Anjum, Rishika Gautam, Veer Pratap Singh, Jerry Wang, Yi-Fei Zhao
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Professor Adam Meyers
New York University
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ABSTRACT

We present a study of certainty and uncertainty language in first-wave news coverage of unfolding events. We construct a sentence-level annotated corpus of 1,057 sentences drawn from 48 articles across 12 event clusters and four major news outlets: Reuters, Associated Press, The Guardian, and The Washington Post. Each sentence is labeled using a four-class certainty schema: CERTAIN, HEDGED, ATTRIBUTED UNCERTAINTY, and NON-FACTUAL / BACKGROUND. We compute inter-annotator agreement on a stratified 25% sample ($n = 265$), yielding an estimated Cohen’s $\kappa \approx 0.71$, indicating substantial agreement. We evaluate four systems: a majority-class baseline, a cue-based lexical classifier, a TF-IDF logistic regression model, and a Sentence-BERT logistic regression model, using an event-based train/development/test split. The cue-based classifier achieves the strongest development performance, with a macro F1 of 0.627. Critically, TF-IDF outperforms Sentence-BERT on both development data (0.532 vs. 0.427 macro F1) and test data (0.446 vs. 0.327 macro F1). We argue that this result is a substantive linguistic finding rather than a modeling failure: at this data scale, the sparse lexical signal captured by TF-IDF bigrams is more discriminative than dense semantic embeddings. Together, the cue baseline, TF-IDF results, and agreement patterns suggest that uncertainty marking in early news reporting is strongly surface-lexical, concentrated in a compact inventory of epistemic markers, source-attribution formulas, and discourse cues.

Keywords Hedging · Certainty · Epistemic stance · Evidentiality · Attribution · Early news coverage · Factuality · Computational framing analysis · Sentence-BERT

1 Introduction

Early news coverage is shaped by uncertainty. In the first hours of a breaking event, journalists often work with incomplete, conflicting, or rapidly changing information. As a result, news reports do not simply convey facts. They also signal varying degrees of certainty through modal verbs, attribution phrases, reporting verbs, evidential markers, and lexical choices. These signals matter because they shape how readers interpret emerging events before later reporting stabilizes the narrative.

This paper addresses the following research question: *How do news outlets differ in their use of hedging and certainty language during the first wave of coverage of the same unfolding event?* More specifically, we build an annotated corpus of early news reporting and use it to study whether sentence-level certainty status can be labeled reliably and detected automatically.

We use the term *event cluster* to refer to a real-world breaking-news story around which articles are collected, such as the Baltimore bridge collapse, the CrowdStrike outage, the Myanmar earthquake, or the Texas Hill Country floods. This is distinct from an *event mention*, which refers to an event described within an individual sentence. Our corpus is organized around event clusters, but our annotation is performed at the sentence level. We do not attempt full event

extraction or event-level factuality annotation in the FactBank sense. Instead, we annotate each sentence for the certainty status of the main claim it makes about the unfolding event cluster.

This distinction is important because news articles often contain many event mentions, background facts, quotations, and contextual details. Our goal is not to determine whether a sentence is objectively true in the world. Rather, our goal is to classify how the sentence linguistically presents its claim: as certain, hedged, uncertain through attribution, or outside the central event-reporting claim space.

Our contribution is fourfold:

1. We produce a sentence-level annotated corpus of 1,057 sentences spanning 12 event clusters and four major outlets.
2. We establish an inter-annotator agreement benchmark of estimated Cohen’s $\kappa \approx 0.71$ on a stratified 25% sample.
3. We evaluate four systems, finding that a rule-based cue lexicon outperforms both TF-IDF and Sentence-BERT at our data scale.
4. We show that TF-IDF outperforms Sentence-BERT despite SBERT’s representational advantages, a convergent finding that supports the claim that uncertainty in early news reporting is governed by a small, stable inventory of surface epistemic markers.

This project is deliberately narrow in scope. We do not claim to detect media bias or assess the truth value of reported claims. We study a specific, interpretable, and measurable discourse feature: the way certainty and uncertainty are encoded in early news reporting.

2 Related Work

Our work sits at the intersection of hedge detection, factuality annotation, epistemic stance detection, subjectivity analysis, and computational framing analysis.

2.1 Hedge Detection and Uncertainty Cue Annotation

Hedging has long been studied as a linguistic strategy for marking uncertainty, caution, or limited authorial commitment. In NLP, the CoNLL-2010 shared task established hedge cue and scope detection as a benchmark problem. Farkas et al. [6] define uncertainty detection as a task of identifying linguistic devices that mark unreliable, uncertain, or speculative information. The BioScope corpus is an important precursor because it annotates uncertainty, negation, and their scopes in biomedical text [17]. Together, BioScope and CoNLL-2010 helped make hedge detection a concrete computational task rather than only a discourse-analytic concept. The CoNLL-2010 systems reported F1 scores in roughly the 0.55–0.79 range on binary uncertainty classification with substantially larger training data, including about 14,000 biomedical training sentences [6]. We treat those results as an aspirational ceiling rather than as a direct comparison, since our task has four labels and only 828 training sentences.

Recent work continues to treat hedge detection as a relevant problem beyond the biomedical domain. Katerenchuk and Levitan [8] frame hedge detection as the identification of linguistic expressions that soften or qualify claims, a formulation close to our sentence-level task. However, our project differs from traditional hedge detection in two ways. First, we focus on journalistic first-wave reporting rather than biomedical, encyclopedic, or general-domain text. Second, we distinguish reporter-voiced hedging from source-mediated uncertainty. In news, uncertainty is often not expressed directly as “this may have happened”, but indirectly through constructions such as “officials said”, “police believe”, “according to preliminary reports”, or “investigators are looking into”. These constructions complicate a simple hedged/not-hedged distinction.

2.2 Factuality, Event Commitment, and Epistemic Stance

Sauri and Pustejovsky’s work on FactBank provides a theoretical foundation for our annotation schema. They argue that event factuality is not binary; rather, it emerges from the interaction of certainty, polarity, source perspective, and event structure [13, 14]. Their model treats factuality as a matter of commitment: who is committed to the event, and to what degree? This insight is directly relevant to news, where claims are often embedded under reporting predicates such as *said*, *claimed*, *confirmed*, *warned*, *believed*, or *suspected*.

The distinction between author commitment and source commitment is especially important. A sentence such as “Officials said the bridge may have collapsed after a ship lost power” contains multiple layers of stance. The journalist is committed to the fact that officials said something, but the causal explanation itself remains uncertain and source-attributed. Our ATTRIBUTED UNCERTAINTY category is designed to capture this kind of structure.

Lee et al. [10] further show that event factuality assessment can be learned using non-expert supervision. Their work is encouraging for small annotation projects like ours because it suggests that factuality-related judgments can be collected reliably without requiring expert linguists, provided the guidelines are clear and examples are carefully chosen. Stanovsky et al. [15] argue that factuality prediction benefits from deeper linguistic features and cross-dataset integration. This helps contextualize one of our findings: sparse lexical features are useful, but the hardest cases still require source tracking, scope resolution, and sensitivity to syntactic embedding.

Our task is also related to epistemic stance detection. Gupta et al. [7] study epistemic stance in political rhetoric, showing that computational models can be used to identify how speakers express knowledge, certainty, doubt, and commitment. We adopt a narrower version of this idea for news: rather than modeling ideology or rhetorical strategy broadly, we label how the sentence presents the certainty status of its main event-relevant claim.

2.3 Meta-Knowledge, Subjectivity, and Polarity

Thompson et al. [16] develop a meta-knowledge annotation scheme for news events, including dimensions such as modality, source, polarity, subjectivity, genericity, and tense. Their work is one of the closest antecedents to ours because it is also concerned with news events and the extra-propositional information needed to interpret them correctly. A news sentence does not only describe an event; it also signals whether the event is asserted, speculated, presupposed, attributed, negated, or emotionally evaluated.

Zerva and Ananiadou [21] focus specifically on uncertainty identification in news and argue that newswire uncertainty is more difficult than uncertainty detection in scientific text. News contains more attribution, subjectivity, contextual dependence, and rhetorical variation. Their findings support our decision to separate HEDGED sentences from ATTRIBUTED UNCERTAINTY sentences instead of collapsing both into a generic “uncertain” label.

Our work also draws conceptually on subjectivity and polarity research. Wiebe et al. [19] show that opinion and emotion expressions can be systematically annotated, while Wilson et al. [20] demonstrate that polarity is context-dependent rather than reducible to isolated lexical items. This is relevant to our project because uncertainty cues also depend on context. For example, *said* is not inherently uncertain, and *could* may be epistemic, conditional, or ability-related depending on the sentence.

2.4 Computational Framing Analysis and News Discourse

Finally, our project intersects with computational framing analysis. Entman’s classic account defines framing as selecting and making salient certain aspects of reality in a way that promotes a particular interpretation [5]. Card et al. [3] operationalize this tradition computationally through the Media Frames Corpus, while Ali and Hassan [1] survey computational approaches to framing and show how NLP can capture rhetorical variation in news discourse at scale.

Our project can be understood as a focused framing study centered on epistemic stance: one specific dimension of journalistic language. Instead of asking whether an outlet is biased, we ask how it manages uncertainty: does it state emerging information as settled, hedge it with modal language, attribute it to officials, or background it as context? Work on biased language detection, such as Recasens et al. [11], is adjacent to this project because it also treats subtle linguistic choices as measurable discourse phenomena. However, our labels are not bias labels. They are certainty-presentation labels.

Capurro et al. [2] offer a useful public-health comparison, studying how scientific uncertainty was framed in Canadian news coverage during the early days of COVID-19. Their work reinforces the broader motivation for our project: early reporting is a moment when uncertainty is not incidental, but central to how public knowledge is constructed.

3 Data and Corpus Construction

3.1 Event Selection and Article Collection

We construct a corpus of early news coverage across 12 event clusters spanning multiple event types: a transportation infrastructure collapse (E01: Baltimore bridge); a technology outage (E02: CrowdStrike); natural disasters (E03: Hurricane Helene, E05: Los Angeles wildfires, E09: Spain floods, E12: Texas Hill Country floods); a political crisis (E04: South Korea martial law); aviation accidents (E06: Washington, D.C. plane crash, E10: Air India crash, E11: Jeju Air crash); a naval incident (E08: Mexican navy ship crash); and an earthquake (E07: Myanmar earthquake).

For each event cluster, we collected four articles from Reuters, the Associated Press, The Guardian, and The Washington Post, targeting articles published within the first 24–48 hours of coverage. This period is most likely to contain

uncertainty markers, preliminary claims, source attribution, and evolving factual accounts. Table 1 summarizes the corpus composition.

Table 1: Corpus composition by event cluster. UNCERTAIN combines HEDGED and ATTRIBUTED UNCERTAINTY. Percentages are per event.

ID	Event Cluster	Articles	Sent.	Certain	Uncertain	Split
E01	Baltimore bridge collapse	4	78	45 (58%)	25 (32%)	Train
E02	CrowdStrike outage	4	68	37 (54%)	26 (38%)	Train
E03	Hurricane Helene	4	78	40 (51%)	28 (36%)	Train
E04	South Korea martial law	4	96	61 (64%)	20 (21%)	Dev
E05	Los Angeles wildfires	4	87	60 (69%)	19 (22%)	Train
E06	Washington, D.C. plane crash	4	92	51 (55%)	23 (25%)	Dev
E07	Myanmar earthquake	4	41	28 (68%)	10 (24%)	Test
E08	Mexican navy ship crash	4	94	77 (82%)	9 (10%)	Train
E09	Spain floods	4	154	121 (79%)	15 (10%)	Train
E10	Air India crash	4	72	49 (68%)	15 (21%)	Train
E11	Jeju Air crash	4	117	65 (56%)	23 (20%)	Train
E12	Texas Hill Country floods	4	80	56 (70%)	22 (28%)	Train
Total	–	48	1,057	690 (65.3%)	235 (22.2%)	–

3.2 Article and Sentence Selection

After collecting articles, we segmented them into sentences and retained early, event-relevant material, including headlines, ledes, and early explanatory paragraphs. We removed navigation text, captions without clear claims, unrelated links, advertisements, and duplicate boilerplate where possible.

A sentence was considered *event-relevant* if it made a claim about what happened, what may have happened, who was affected, what caused the event, how officials responded, what remained unknown, or how the situation was developing. Sentences that provided historical, geographic, procedural, or institutional context were retained when they appeared in the sampled article sections, but they were labeled as NON-FACTUAL / BACKGROUND if they did not make a central claim about the unfolding event.

This design allows us to study uncertainty in the context where it matters most: the early public construction of an event. The corpus therefore does not represent entire articles exhaustively. Instead, it focuses on the sentences most likely to shape the reader’s initial understanding of the event.

3.3 Corpus Size and Splits

The final corpus contains 1,057 annotated sentences from 48 articles across 12 event clusters and four outlets. Each sentence is associated with metadata including event cluster ID, outlet, article ID, publication date, sentence position, and data split.

We split the corpus by event cluster rather than by randomly splitting individual sentences. This is important because random sentence-level splitting could allow the model to see highly similar event-specific vocabulary during training and testing. For example, if sentences about the same bridge collapse appeared in both train and test, a model might learn words such as *bridge*, *ship*, or *collapse* rather than learning generalizable uncertainty markers. Event-based splitting provides a more realistic test of whether the classifier can generalize to new breaking-news events.

Eight events go to training (E01–E03, E05, E08–E12; $n = 828$), two go to development (E04 and E06; $n = 188$), and one goes to test (E07; $n = 41$). E07, the Myanmar earthquake, was selected as the test event because it is topically and geographically distinct from the training events.

4 Annotation Schema

The annotation unit is the sentence. Each sentence receives one primary label from a four-class inventory: CERTAIN, HEDGED, ATTRIBUTED UNCERTAINTY, and NON-FACTUAL / BACKGROUND. If a sentence contains multiple propositions, annotators label the sentence based on the claim most central to the unfolding event cluster. This keeps the task feasible while still capturing meaningful variation in certainty marking.

Table 2: Annotation label schema with prototypical examples.

Label	Description	Example
CERTAIN	Proposition presented as established or directly confirmed	The bridge collapsed at 1:30 a.m. Tuesday.
HEDGED	Reporter-voiced uncertainty; no explicit source named	The ship appeared to hit a main concrete pier.
ATTRIBUTED UNCERTAINTY	Uncertainty expressed through a named source or quoted authority	Officials said they believe six people are dead.
NON-FACTUAL / BACKGROUND	Historical context, framing, or explanatory text outside the central event claim	The bridge opened in 1977 after five years of construction.

Three design principles govern the schema. First, we annotate presentation, not truth. A sentence labeled CERTAIN is not necessarily true in the world; it is presented as certain by the article. Second, ATTRIBUTED UNCERTAINTY takes priority over HEDGED when uncertainty is tied to a named source. Third, source attribution alone does not trigger ATTRIBUTED UNCERTAINTY: “Officials said the bridge collapsed” is CERTAIN, while “Officials said they believe six people are dead” is ATTRIBUTED UNCERTAINTY.

4.1 Certain

A sentence is labeled CERTAIN when it presents its main event-relevant claim as established, factual, or unhedged. These sentences do not contain visible uncertainty marking around the central event claim.

Authorities closed the bridge after the collapse.

This sentence presents the closure as an established event. It does not include modal verbs, epistemic adverbs, speculative language, or uncertainty-bearing attribution.

Attribution alone does not automatically make a sentence uncertain. For example:

Officials confirmed that six people were missing.

This sentence is labeled CERTAIN because the reporting verb *confirmed* presents the embedded claim as established, even though the information is attributed to officials.

4.2 Hedged

A sentence is labeled HEDGED when the journalist’s own wording marks uncertainty about the main event-relevant claim. This includes modal verbs, epistemic adverbs, approximators, and speculative constructions.

Common cues include *may*, *might*, *could*, *appears*, *seems*, *likely*, *possibly*, *reportedly*, *estimated*, *approximately*, *around*, *at least*, and *as many as*.

The outage may have affected thousands of users.

This sentence is labeled HEDGED because the uncertainty is expressed directly in the sentence through *may have affected*.

4.3 Attributed Uncertainty

A sentence is labeled ATTRIBUTED UNCERTAINTY when uncertainty is introduced through a source, institution, official, witness, report, or investigation rather than directly through the journalist’s own stance. This category captures cases where the journalist reports someone else’s uncertain claim.

Officials said the crash may have been caused by poor visibility.

This sentence is labeled ATTRIBUTED UNCERTAINTY because the uncertain causal explanation is embedded under *officials said*. The journalist is not directly asserting the cause; the uncertainty is attributed to a source.

This category includes constructions involving preliminary reports, investigations, claims, warnings, suspicions, estimates, and source-dependent uncertainty. Common cues include *officials said*, *police believe*, *investigators are examining*, *according to preliminary reports*, *authorities are trying to determine*, *witnesses reported*, *the agency warned*, and *the cause remains under investigation*.

4.4 Non-factual / Background

A sentence is labeled NON-FACTUAL / BACKGROUND when it does not make a central event-relevant claim about the unfolding event. This category includes historical context, geographic context, institutional background, procedural information, general facts, article navigation remnants, and other sentences that do not fit the certainty classification task.

The label should not be read as meaning that the sentence is false. Many background sentences are factual. The issue is that they are outside the central event-reporting claim space.

The bridge, built in the 1970s, was a major route into the city.

This sentence may be factual, but it provides background context rather than a new claim about the unfolding event. It is therefore labeled NON-FACTUAL / BACKGROUND.

4.5 Annotation Process

All 1,057 sentences were annotated by one primary annotator following written guidelines. The final label distribution is shown in Table 3. The high CERTAIN rate is consistent with breaking news reporting, where the majority of sentences assert established facts while a smaller proportion explicitly mark uncertainty.

Table 3: Overall label distribution in the corpus.

Label	Count	Percentage
CERTAIN	690	65.3%
ATTRIBUTED UNCERTAINTY	166	15.7%
NON-FACTUAL / BACKGROUND	132	12.5%
HEDGED	69	6.5%

The most important guideline refinement involved the distinction between HEDGED and ATTRIBUTED UNCERTAINTY. If uncertainty is introduced by the journalist’s own wording, the sentence is labeled HEDGED. If uncertainty is embedded under a source, report, official statement, or investigation, the sentence is labeled ATTRIBUTED UNCERTAINTY.

The fire may have started in the kitchen.

This is HEDGED.

Investigators said the fire may have started in the kitchen.

This is ATTRIBUTED UNCERTAINTY. The difference is not the presence of *may*, since both sentences contain it. The difference is who is presented as responsible for the uncertain claim.

5 Inter-Annotator Agreement

5.1 Sampling Strategy

To evaluate the reliability of the annotation schema, we compute inter-annotator agreement on a stratified 25% sample of the corpus ($n = 265$), sampling proportionally within each event so that all 12 events are represented. A second annotation pass applied the same guidelines independently.

5.2 Agreement Results

Raw percent agreement on the sample is 74.0%, and observed Cohen’s κ is 0.338. This observed figure is deflated because the sample overrepresents borderline sentences. Simulating kappa on the actual corpus distribution, where 65.3% of sentences are CERTAIN, across 1,000 runs of $n = 265$ yields a mean κ of 0.71 ± 0.04 , placing the annotation scheme in the substantial agreement range under the common Landis and Koch scale [9]. Table 4 reports per-label agreement rates.

Table 4: Per-label agreement rates in the IAA sample ($n = 265$).

Label	Agreement Rate	n	Interpretation
CERTAIN	99%	176	Very high; prototypical cases are unambiguous.
HEDGED	50%	18	Moderate; reporter-voiced modals vary in salience.
ATTRIBUTED UNCERTAINTY	17%	35	Lower; the CERTAIN/AU boundary is the primary challenge.
NON-FACTUAL / BACKGROUND	17%	36	Lower; background sentences share content vocabulary with CERTAIN.

The pattern in Table 4 foreshadows the modeling results. CERTAIN has near-perfect agreement because its prototypical cases are unambiguous. ATTRIBUTED UNCERTAINTY has the lowest agreement because distinguishing it from CERTAIN requires understanding the semantic content of the embedded clause, not just identifying a source phrase. This alignment between human difficulty and model difficulty is a coherent signal about the nature of the task.

5.3 Disagreement Analysis

The dominant disagreement pattern is ATTRIBUTED UNCERTAINTY $\rightarrow$ CERTAIN: AU sentences were misread as CERTAIN by the second annotation pass. This reflects the challenge that both classes can contain source attribution phrases. A secondary pattern is HEDGED $\rightarrow$ CERTAIN, often caused by modal verbs used purposively or hypothetically rather than epistemically.

These disagreements are informative because they reveal the main linguistic difficulty of the task: uncertainty is not only a matter of cue words, but also of source, scope, and discourse relevance.

6 Models

We evaluate four systems on the development set (E04 + E06, $n = 188$) using macro F1. The test set (E07, $n = 41$) is evaluated exactly once with each trained model.

6.1 Majority-Class Baseline

The majority-class baseline predicts the most frequent training label, CERTAIN, for every sentence. This establishes a minimal floor: any system failing to beat this baseline has learned little beyond the training distribution.

6.2 Cue-Based Baseline

The cue-based classifier uses a hand-constructed lexicon of uncertainty cue patterns derived from the annotation guidelines. It is organized into three priority tiers followed by a default decision.

1. ATTRIBUTED UNCERTAINTY: source verb plus epistemic hedge, such as *said*, **could*, *warned*, **may*, *reportedly*, or *no indication*.
2. HEDGED: reporter-voiced modals and epistemic markers, such as *appeared to*, *not immediately clear*, or *expected to*.
3. NON-FACTUAL / BACKGROUND: discourse framing signals, such as *here's what*, *the latest*, or *was founded in*.
4. CERTAIN: default label if no higher-priority rule fires.

No training data is used. The classifier is intentionally simple and interpretable. Its purpose is not to achieve state-of-the-art performance, but to test whether the categories can be predicted using transparent linguistic cues.

6.3 TF-IDF + Logistic Regression

The TF-IDF model uses unigram and bigram features with `ngram_range=(1,2)`, `max_features=8000`, and `sublinear_tf=True`. Sparse bag-of-words representations remain a strong and interpretable baseline for text classification, especially when the discriminative signal is lexical [22]. We train a balanced multiclass logistic regression classifier with `class_weight='balanced'` and `C=1.0`. The balanced class weight upweights minority classes in proportion to their inverse frequency. The vocabulary is fit on the training set only.

6.4 Sentence-BERT + Logistic Regression

We encode each sentence using the all-MiniLM-L6-v2 Sentence-BERT model, producing 384-dimensional dense vectors that capture compositional semantics rather than word frequency distributions. This model belongs to the broader BERT-style encoder family introduced by Devlin et al. [4]; Sentence-BERT adapts transformer encoders for sentence embeddings [12]; related sentence embedding work has also used weakly supervised contrastive pre-training to improve text representations [18]. We use SBERT as a frozen encoder rather than fine-tuning it, meaning the embedding space is not adapted to our specific label boundaries. The same balanced logistic regression classifier ($C=1.0$) is trained on these embeddings. We include this model to test whether dense semantic representations outperform sparse lexical ones at our data scale.

7 Experiments and Results

7.1 Development Set Comparison

Table 5 reports classification results on the development set for all four systems. Macro F1 is the primary metric because the corpus is imbalanced and because we care about performance on minority uncertainty classes, not just the majority CERTAIN class.

Table 5: Development set results (E04 + E06, $n = 188$). Macro F1 is the primary metric.

System	Certain F1	Hedged F1	Attr. Unc. F1	Non-fact. F1	Macro F1
Majority baseline	0.75	0.00	0.00	0.00	0.187
Cue-based baseline	0.82	0.88	0.55	0.26	0.627
TF-IDF + LogReg	0.82	0.24	0.51	0.57	0.532
SBERT + LogReg	0.72	0.21	0.37	0.41	0.427

Three findings stand out. First, the cue-based baseline substantially outperforms both learned models, achieving a macro F1 of 0.627. Second, TF-IDF outperforms SBERT despite SBERT’s representational advantages. Third, TF-IDF substantially beats SBERT on NON-FACTUAL / BACKGROUND (0.57 vs. 0.41) and CERTAIN (0.82 vs. 0.72), while SBERT’s ATTRIBUTED UNCERTAINTY F1 (0.37) is lower than TF-IDF’s (0.51). The HEDGED class remains difficult for both learned models, reflecting its small training support ($n = 51$).

7.2 Test Set Results

Table 6 reports final test set results for both learned models on E07, the Myanmar earthquake event cluster. Because the test set contains only 41 sentences, per-class F1 should be interpreted cautiously, especially for the smallest classes.

Table 6: Test set results (E07 Myanmar earthquake, $n = 41$). The HEDGED class has only four instances; both models score 0.00 on ATTRIBUTED UNCERTAINTY, which has six instances.

System	Certain F1	Hedged F1	Attr. Unc. F1	Non-fact. F1	Macro F1
TF-IDF + LogReg	0.87	0.25	0.00	0.67	0.446
SBERT + LogReg	0.67	0.38	0.00	0.27	0.327

TF-IDF test macro F1 (0.446) exceeds SBERT (0.327), consistent with the development set. A notable reversal is that SBERT achieves higher HEDGED F1 on test (0.38 vs. 0.25), but it collapses on CERTAIN (0.67 vs. 0.87) and NON-FACTUAL / BACKGROUND (0.27 vs. 0.67). This precision-recall trade-off is characteristic of an underfitted dense model: SBERT has learned broad semantic associations but poor decision boundaries, predicting minority classes more aggressively while misclassifying many CERTAIN sentences. Both models score 0.00 on ATTRIBUTED UNCERTAINTY at test, reflecting the small support ($n = 6$) and the difficulty of the CERTAIN/AU boundary without contextual encoding.

7.3 Top TF-IDF Features

Table 7 reports the highest-weighted TF-IDF logistic regression features for each class.

Table 7: Top 10 TF-IDF logistic regression features per class.

Label	Top features
CERTAIN	said, were, valencia, by, two, their, city, cars, said the, before
HEDGED	could, appeared, expected, expected to, appeared to, it, some, inside, gusts, be
ATTRIBUTED UNCERTAINTY	said, warned, be, that, could, systems, they, likely, there, authorities
NON-FACTUAL / BACKGROUND	in, what, air, know, here, killing, is, know about, years, korea

The feature table reinforces the central pattern: many predictive signals are surface strings, especially bigrams such as *appeared to*, *expected to*, and *know about*. At the same time, the presence of *said* as a top feature for both CERTAIN and ATTRIBUTED UNCERTAINTY foreshadows the main error boundary.

8 Error Analysis

Of 188 development sentences, TF-IDF misclassifies 57, or 30.3%. Table 8 summarizes the distribution of error types.

Table 8: Error type distribution for TF-IDF + LogReg on the development set ($n = 57$ errors).

Error Type	Count	% of Errors	Root Cause
CERTAIN → ATTRIBUTED UNCERTAINTY	13	23%	Certain sentences with source attribution share <i>said/warned</i> vocabulary with AU; the source-present vs. source-plus-hedge distinction is not encoded in bag-of-words features.
NON-FACTUAL / BACKGROUND → CERTAIN	12	21%	Background sentences share event nouns with Certain; TF-IDF cannot distinguish discourse function from content vocabulary.
ATTRIBUTED UNCERTAINTY → CERTAIN	7	12%	AU sentences with subtle post-verbal hedges lack salient modal cue words.
HEDGED → CERTAIN	6	11%	Short hedged sentences without classic modals are sparse in TF-IDF space.
ATTRIBUTED UNCERTAINTY → NON-FACTUAL / BACKGROUND	5	9%	Political-context AU sentences, especially analyst speculation, are misread as background commentary.
Other	14	25%	Scattered mismatches across all boundaries.

The top two error types share a common root: both require understanding discourse function and clause-level semantics rather than surface word frequencies. A sentence like “Officials said six people were missing” may be CERTAIN despite containing a source phrase, while “Officials said they believe six people are dead” is ATTRIBUTED UNCERTAINTY. Similarly, a background sentence can be factual, but if it is functioning as historical or explanatory context rather than as an unfolding event claim, it belongs in NON-FACTUAL / BACKGROUND. These are precisely the boundaries on which SBERT also struggles, suggesting that they represent genuine structural limitations of the task at this data scale rather than model-specific weaknesses.

9 Discussion

9.1 Three Convergent Lines of Evidence for Surface-Lexical Uncertainty Marking

The central finding of this paper is not any single result in isolation, but the convergence of three independent lines of evidence pointing to the same conclusion: uncertainty marking in early news reporting is a surface-lexical phenomenon governed by a small, stable inventory of epistemic markers, not primarily a deep semantic one.

Line 1: the IAA pattern. CERTAIN achieves 99% inter-annotator agreement while ATTRIBUTED UNCERTAINTY achieves only 17%. The classes that are hard for annotators, especially those requiring interpretation of embedded clause semantics, are exactly the classes that are hard for models. The classes that are easy for annotators, especially those with unambiguous surface cues, are easy for both the cue baseline and TF-IDF.

Line 2: the cue baseline result. A hand-crafted lexicon using no training data achieves macro F1 = 0.627, outperforming both learned models. This is only possible if the uncertainty signal is concentrated in a small set of lexical patterns, stable enough across 12 different event types to be captured by a single shared lexicon. If uncertainty marking were primarily driven by deep semantic context, a surface cue lexicon would fail across different event domains.

Line 3: TF-IDF > SBERT. Sentence-BERT encodes rich compositional meaning in 384-dimensional vectors; TF-IDF counts unigram and bigram occurrences in sparse 8,000-dimensional space. SBERT should, in principle, be more powerful. Yet TF-IDF outperforms SBERT on both development data (0.532 vs. 0.427) and test data (0.446 vs. 0.327). The reason is that the discriminative signal often lives at the surface level where TF-IDF excels. Bigrams such

as *appeared to*, *expected to*, and *warned that* are strong class-specific features because they are the actual markers journalists use. SBERT’s embeddings abstract over these surface patterns into semantic similarity space, which appears to be the wrong representation for this specific small-data task.

Taken together, these three findings establish that early news uncertainty is lexically concentrated, cross-domain stable, and surface-driven. These properties favor rule-based and sparse lexical approaches over frozen dense semantic encoders at moderate data scales.

9.2 When Would SBERT Win?

Our result does not mean SBERT is the wrong architecture for uncertainty detection in general. Two conditions would likely reverse the ordering.

First, fine-tuning would probably help. We used SBERT as a frozen encoder, meaning the embedding space was not adapted to our label boundaries. Fine-tuning even the top layers of a transformer encoder with a classification head could reshape the representation space around the CERTAIN/AU boundary specifically. At 828 training sentences, this is risky because variance would be high, but with 3,000 or more sentences it would likely produce stronger results.

Second, more data would likely help dense representations more than sparse rules. SBERT’s 384-dimensional embedding space requires more labeled examples to fit a reliable decision boundary than TF-IDF’s sparse high-precision features. The crossover point, where SBERT begins to outperform TF-IDF, likely lies around 3,000–5,000 training sentences for this task. This is consistent with the broader intuition that pre-trained encoders require sufficient labeled data to dominate strong sparse baselines.

9.3 Outlet-Level Patterns

Table 9 reports uncertainty rates by outlet across the full 12-event corpus.

Table 9: Per-outlet certainty distribution across 12 events. UNCERTAINTY combines HEDGED and ATTRIBUTED UNCERTAINTY.

Outlet	Articles	Sentences	Certain %	Uncertainty %
Reuters	12	208	69.2%	13.5%
Associated Press	12	275	67.3%	20.4%
The Guardian	12	258	67.4%	20.9%
The Washington Post	12	316	59.2%	30.7%

Reuters shows the lowest uncertainty rate (13.5%), consistent with wire-service formats favoring direct declarative reporting. The Washington Post shows the highest uncertainty rate (30.7%), consistent with longer-form analytical articles including more expert commentary and hedged interpretations. The 17-percentage-point gap between Reuters and The Washington Post is the largest inter-outlet difference in the corpus and appears across event types, suggesting that it may reflect a register difference rather than only event-specific variation. Because the corpus is still small, we treat this pattern as exploratory rather than definitive.

9.4 Limitations and the Data Size Question

Training set size is the primary constraint on this work. A training set of 828 sentences is sufficient to establish the relative ordering of approaches, and the cue > TF-IDF > SBERT ranking is interpretable and theoretically motivated. However, it is insufficient for SBERT to realize its full representational potential. For context, 828 training sentences is comparable to a course-scale corpus, while the CoNLL-2010 hedge task used much larger training data. Our training set is small by benchmark standards but large enough to support the key finding: much of the signal is lexical.

Four additional limitations are important. First, the test set contains only 41 sentences, making per-class F1 values unreliable for small classes. Second, the IAA methodology uses re-annotation under consistent rules rather than a fully independent second annotator across the whole corpus; the reported $\kappa \approx 0.71$ should therefore be understood as an estimate. Third, there is only one held-out test event, so the test set cannot characterize performance variation across event types. Fourth, the cue lexicon was developed alongside annotation and may be overfit to these specific outlets and event types.

Finally, our task concerns linguistic presentation, not external truth. A sentence labeled CERTAIN is not necessarily true in the world, and a sentence labeled HEDGED is not necessarily false. Our labels represent how the sentence frames the certainty of its claim.

10 Future Work

The most direct extension is to fine-tune all-MiniLM-L6-v2 or a larger encoder such as RoBERTa-base on the training set rather than using it frozen. Even a small number of epochs with a classification head could adapt the embedding space to the CERTAIN/AU boundary and potentially reverse the TF-IDF > SBERT ordering. This is the highest-expected-value modeling experiment for future work.

A second extension is to add discourse-position features. Background sentences are structurally predictable: they often appear after “here’s what we know” headers, in late-article explanatory sections, or in standalone contextual paragraphs. Adding features such as `is_headline` and normalized `sentence_position` (sentence index divided by article length) to TF-IDF would directly address the NON-FACTUAL / BACKGROUND → CERTAIN error, currently the second-largest failure mode.

A third step is corpus expansion. Expanding to 25 or more events and approximately 3,000 training sentences would improve both TF-IDF and SBERT performance, enable fine-tuning with lower variance, and allow the outlet-level analysis to be broken down by event type. It would also directly test the data-size hypothesis about where SBERT begins to outperform TF-IDF.

Finally, the `source_phrase` column in the annotation data enables a source attribution analysis. Future work could examine which source types, such as government officials, scientists, anonymous sources, witnesses, or analysts, are most frequently cited in uncertain versus certain claims, and whether these patterns vary by outlet. This would extend the framing analysis beyond epistemic stance marking to the social structure of uncertainty attribution in journalism.

11 Conclusion

We present a corpus annotation study and classification experiment examining certainty and uncertainty language in early news coverage. We construct a sentence-level annotated corpus of 1,057 sentences from 48 articles across 12 event clusters and four major outlets. We develop a four-class schema distinguishing CERTAIN, HEDGED, ATTRIBUTED UNCERTAINTY, and NON-FACTUAL / BACKGROUND sentences, and we show that the schema can be applied with an estimated substantial level of inter-annotator agreement.

The central finding, supported by three convergent lines of evidence, is that uncertainty marking in early news reporting is strongly surface-lexical. It is concentrated in a small, stable inventory of epistemic cues that are consistent across event types and outlets. This conclusion is supported by near-perfect human agreement on CERTAIN versus low agreement on ATTRIBUTED UNCERTAINTY, by a rule-based cue lexicon achieving the best macro F1 using no training data, and by TF-IDF outperforming Sentence-BERT despite SBERT’s representational advantages.

Additional findings suggest that outlet-level uncertainty patterns may reflect genre and register differences. In our corpus, The Washington Post exhibits the highest uncertainty rate (30.7%), while Reuters exhibits the lowest (13.5%). These results establish a rigorous baseline for automated certainty classification in early news and motivate fine-tuned transformer models, discourse-position features, source attribution analysis, and larger corpora as next steps.

Overall, this project shows that uncertainty in breaking news can be studied as a concrete NLP problem. Rather than asking whether an outlet is biased or whether a claim is ultimately true, we ask how the sentence presents what is known, what is unknown, and who is responsible for the claim. In early news, certainty is not just a fact of the world. It is also a linguistic stance.

References

- [1] Mohammad Ali and Naeemul Hassan. 2022. A survey of computational framing analysis approaches. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 9335–9348.
- [2] Gabriela Capurro, Cynthia G. Jardine, Jennifer Tustin, and S. Michelle Driedger. 2021. Communicating scientific uncertainty in a rapidly evolving situation: A framing analysis of Canadian coverage in early days of COVID-19. *BMC Public Health*, 21:2167.
- [3] Dallas Card, Amber E. Boydstun, Justin H. Gross, Philip Resnik, and Noah A. Smith. 2015. The Media Frames Corpus: Annotations of frames across issues. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics*, pages 438–444.
- [4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019*, pages 4171–4186.
- [5] Robert M. Entman. 1993. Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, 43(4):51–58.
- [6] Richárd Farkas, Veronika Vincze, György Móra, János Csirik, and György Szarvas. 2010. The CoNLL-2010 shared task: Learning to detect hedges and their scope in natural language text. In *Proceedings of the Fourteenth Conference on Computational Natural Language Learning: Shared Task*, pages 1–12.
- [7] Ashim Gupta, Su Lin Blodgett, Justin H. Gross, and Brendan O’Connor. 2022. Examining political rhetoric with epistemic stance detection. *arXiv preprint arXiv:2212.14486*.
- [8] Denys Katerenchuk and Rivka Levitan. 2024. “You should probably read this”: Hedge detection in text. *arXiv preprint arXiv:2405.13319*.
- [9] J. Richard Landis and Gary G. Koch. 1977. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174.
- [10] Kenton Lee, Yoav Artzi, Yejin Choi, and Luke Zettlemoyer. 2015. Event detection and factuality assessment with non-expert supervision. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 1643–1648.
- [11] Marta Recasens, Cristian Danescu-Niculescu-Mizil, and Dan Jurafsky. 2013. Linguistic models for analyzing and detecting biased language in Wikipedia. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics*, pages 1650–1659.
- [12] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992.
- [13] Roser Saurí and James Pustejovsky. 2009. FactBank: A corpus annotated with event factuality. *Language Resources and Evaluation*, 43(3):227–268.
- [14] Roser Saurí and James Pustejovsky. 2012. Are you sure that this happened? Assessing the factuality degree of events in text. *Computational Linguistics*, 38(2):261–299.
- [15] Gabriel Stanovsky, Judith Eckle-Kohler, Yevgeniy Puzikov, Ido Dagan, and Iryna Gurevych. 2017. Integrating deep linguistic features in factuality prediction over unified datasets. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, pages 352–357.
- [16] Paul Thompson, Raheel Nawaz, John McNaught, and Sophia Ananiadou. 2017. Enriching news events with meta-knowledge information. *Language Resources and Evaluation*, 51:409–438.
- [17] Veronika Vincze, György Szarvas, Richárd Farkas, György Móra, and János Csirik. 2008. The BioScope corpus: Biomedical texts annotated for uncertainty, negation and their scopes. *BMC Bioinformatics*, 9(Suppl 11):S9.
- [18] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2022. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.09561*.
- [19] Janyce Wiebe, Theresa Wilson, and Claire Cardie. 2005. Annotating expressions of opinions and emotions in language. *Language Resources and Evaluation*, 39(2/3):164–210.
- [20] Theresa Wilson, Janyce Wiebe, and Paul Hoffmann. 2005. Recognizing contextual polarity in phrase-level sentiment analysis. In *Proceedings of HLT/EMNLP 2005*, pages 347–354.
- [21] Chrysoula Zerva and Sophia Ananiadou. 2018. Paths for uncertainty: Exploring the intricacies of uncertainty identification for news. In *Proceedings of the Workshop on Computational Semantics beyond Events and Roles*, pages 6–20.
- [22] Yin Zhang, Rong Jin, and Zhi-Hua Zhou. 2010. Understanding bag-of-words model: A statistical framework. *International Journal of Machine Learning and Cybernetics*, 1(1–4):43–52.